RF-Lab · AD9910 DDS signal generator

RF-Lab Manual

Signal generator, sweeper and level meter
on an Arduino Mega with the AD9910 shield, run from a PC

Firmware 1.0 · As of 2026-11-03

RF-Lab turns an Arduino Mega 2560 with the GRA & AFCH AD9910 DDS shield into a small RF laboratory operated from a Windows PC: a signal generator with AM, FM and sweeps, a scalar response analyser, a level meter and three clock outputs.



github.com/andiepp/RF-Lab

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1 Introduction

1.1 What RF-Lab is

The board generates the signals; the PC application sets them. The board keeps no readable signal state: the application holds every value shown in its window and sends the whole setting whenever one value changes. The board reports what it can report: its identity, whether its clock generator is fitted, its LED and its detector input.

RF-Lab is the standalone form of the RF-MiniLab instrument of the HASE laboratory system. Board and firmware are the same: a board flashed for one works with the other without reflashing (see "Using the board with HASE").

1.2 The hardware

Part	Purpose
Arduino Mega 2560	controller; USB serial port to the PC (115200 baud)
AD9910 DDS V3 shield (GRA & AFCH)	the signal: 100 kHz – 300 MHz, 50 MHz TCXO, AM, FM and sweeps in the DDS itself
Si5351 clock generator (I^2C)	three square-wave clock outputs, 10 kHz – 160 MHz
AD8307 log detector, 50 Ω input	the level meter: about -70 to +10 dB, read by the Mega's converter (input A8, 2.56 V reference)
128×64 OLED (I^2C)	shows the board's own state
+5 V probe output	supply for high-impedance probes ahead of the detector

The signal leaves the board at the shield's RF output. Connect the device under test between the RF output and the detector input to measure a response (ANALYZE), or the detector alone to measure a level (MEASURE).

1.3 Ranges

Setting	Range
Carrier frequency	100 kHz – 300 MHz, 1 Hz steps
Attenuation	0 – 75 dB below full scale
AM	modulation 1 – 99 999 Hz, depth 0 – 99 %
FM	modulation 1 – 99 999 Hz, deviation 1 – 999 999 Hz
Sweep	start and stop 100 kHz – 300 MHz, 1 – 16 000 ms
ANALYZE	10 – 500 points
MEASURE	one reading every 10 – 16 000 ms
Clock outputs	10 kHz – 160 MHz each

2 Installation

2.1 The application

Unpack `RF-Lab-<version>-windows-x64.zip` anywhere and start `RfLab.exe`. It is a single program for Windows 10 and 11 (x64) and needs no .NET installation. It keeps its own files in `%LocalAppData%\RF-Lab`:

File	Contents
`preferences.txt`	the theme and the last serial port
`Presets\`	the stored settings; the examples from the zip's `presets` folder are copied here at the first start
`RF-Lab Manual.pdf`	this manual, written out when **Manual** is clicked

To update, replace `RfLab.exe`; the stored settings stay.

2.2 The USB driver

The Mega's USB serial port appears as a COM port. Boards with the CH340 USB chip (most Mega clones) need the CH340 driver, which Windows Update usually installs by itself. The COM number belongs to the USB socket: plug the board into another socket and it gets another number.

2.3 The firmware

A board that already serves the HASE RF-MiniLab runs the right firmware. Otherwise write it with the flash tool from the zip's `firmware` folder:

1 Close the RF-Lab application (and a HASE host using the board): the

port must be free.

1 Double-click `firmware\Flash-Firmware.cmd` and enter the board's COM

port, or in PowerShell:

```
.\Flash-Firmware.ps1 -Port COM16
```

The tool uses avrdude, which comes with the Arduino IDE (it finds it in `%LocalAppData%\Arduino15`); name another one with `-Avrdude`. `-Check` only asks the bootloader for the chip's signature and writes nothing. Building the firmware from source is described in `docs/Building.md`.

3
The window



The window in the Classic theme, measuring with the detector

3.1 Top bar

- **Port**: the serial ports present; □ looks for them again (after

plugging the board in).

- **Connect** opens the port. The Mega restarts when its port opens, so

connecting takes about three seconds. The application then checks that an RF-Lab board answers (its node type AE.70.10.80), reads whether the clock generator is fitted and the LED's state, and sends the window's setting — the carrier or modulation and the three clock frequencies — because the board restarted on its own defaults.

- **Disconnect** closes the port. The board keeps generating its last

signal; there is no "output off" command.

- The lamp and text beside the keys show the connection.
- **Theme** switches between Classic, Studio dark and Studio light at once,

also while connected.

- **Manual** (or F1) opens this manual.

3.2 Signal

The **Frequency** display shows the carrier in Hz, grouped MHz · kHz · Hz; the **A** display shows the attenuation in dB below full scale. Change a digit with the mouse wheel over it, or with the small keys that appear above and below it under the mouse; a carry runs into the next digit. Or type a number into the field in the display's title and press Enter.

Every change is sent at once in OFF, AM and FM. While a dial turns, the application sends one request at a time and always the newest value.

3.3 Modulation and measurement

The mode keys on the left choose what the board does: **OFF** (an unmodulated carrier — "OFF" means modulation off), **AM**, **FM**, **SWEEP**, **ANALYZE** and **MEASURE**. The fields beside them are enabled in the modes they belong to. The next chapter describes each mode.

3.4 Measurements, clock outputs, trace

The tabs in the middle show the chart of the measured values with the current value as a bar on its left, the clock outputs (see "Clock outputs") and the trace of the board's messages (see "The trace").

3.5 Board and stored settings

The right column shows the connected board: port, node type, message buffer and whether the Si5351 clock generator is fitted. Below it are the stored settings (see "Stored settings").

3.6 Status line

The status line reports the last action ("AM: 10 MHz, -20 dB, 1000 Hz, 80 %.") or the last error; an error also lights the red strip at the window's left edge. The **LED** lamp at the right shows the board's LED as the board reports it; click it to switch the LED. The LED is a quick test that the board is listening: it answers with its real state.

4
Operating modes

4.1 OFF: the carrier

The board generates the frequency and attenuation of the Signal displays. Any change is sent at once.

4.2 AM

Amplitude modulation of the carrier with a sine of **F mod** Hz and a depth of **AM [%]**. The firmware computes the modulation into the AD9910's RAM, which takes a moment; the AD9910 then plays it by itself.

4.3 FM

Frequency modulation with a sine of **F mod** Hz and a deviation of **FM dev** Hz, computed into the AD9910's RAM like AM.

4.4 SWEEP

The AD9910's own frequency sweep from **F start** to **F stop** in **T sweep** ms, at the Signal attenuation. **Mode** chooses its form:

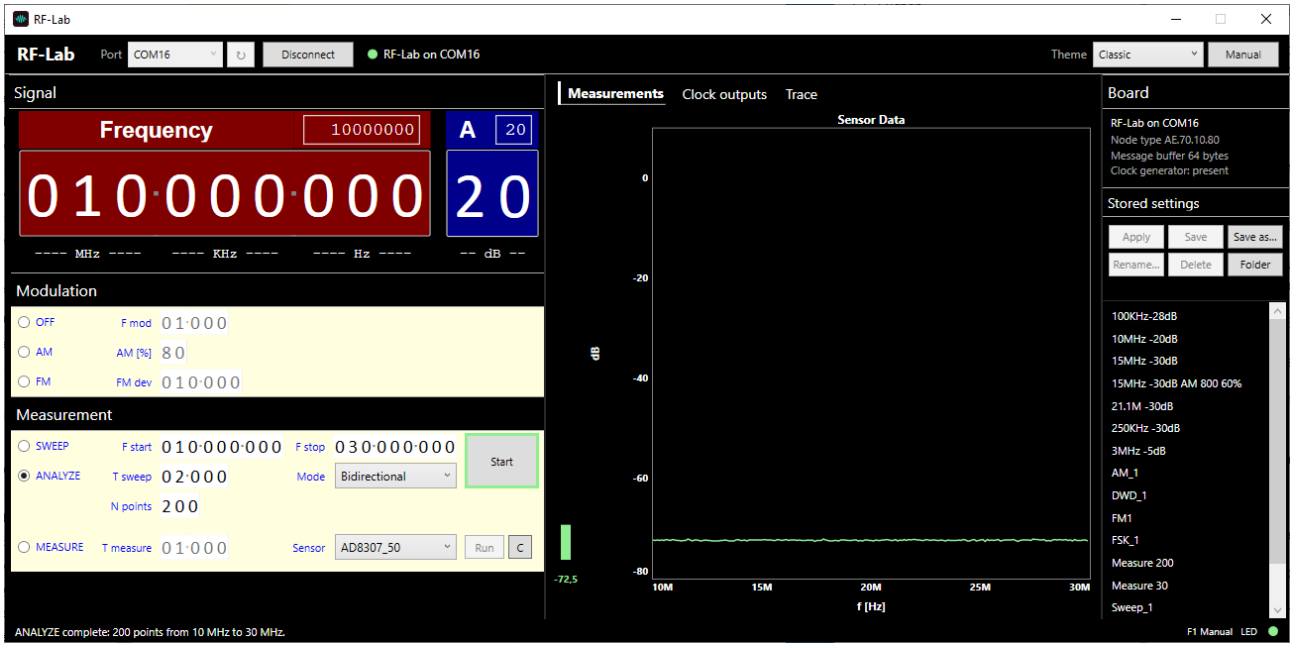
Mode	Form
Bidirectional	up and down again, a triangle
Ramp	up, then back to the start at once, a sawtooth
SingleRamp	one ramp up; the frequency stays at the stop

Start sends the sweep; the board then sweeps by itself until it gets another setting. **Stop** sends the carrier again. Choosing SWEEP sends nothing until Start.

4.5 ANALYZE

A scalar response over frequency: the application steps the carrier from **F start** to **F stop** in **N points** steps, waits for the detector to settle, reads it and plots the point. Connect the device under test between the RF output and the detector.

The settling time per point is **T sweep** divided by the points, with the original application's floor of 3.2 s for the whole run (the round trip of a step counts towards it). **Start** begins a run, **Stop** ends it early; either way the carrier returns to the Signal frequency afterwards.



ANALYZE, 200 points from 10 to 30 MHz

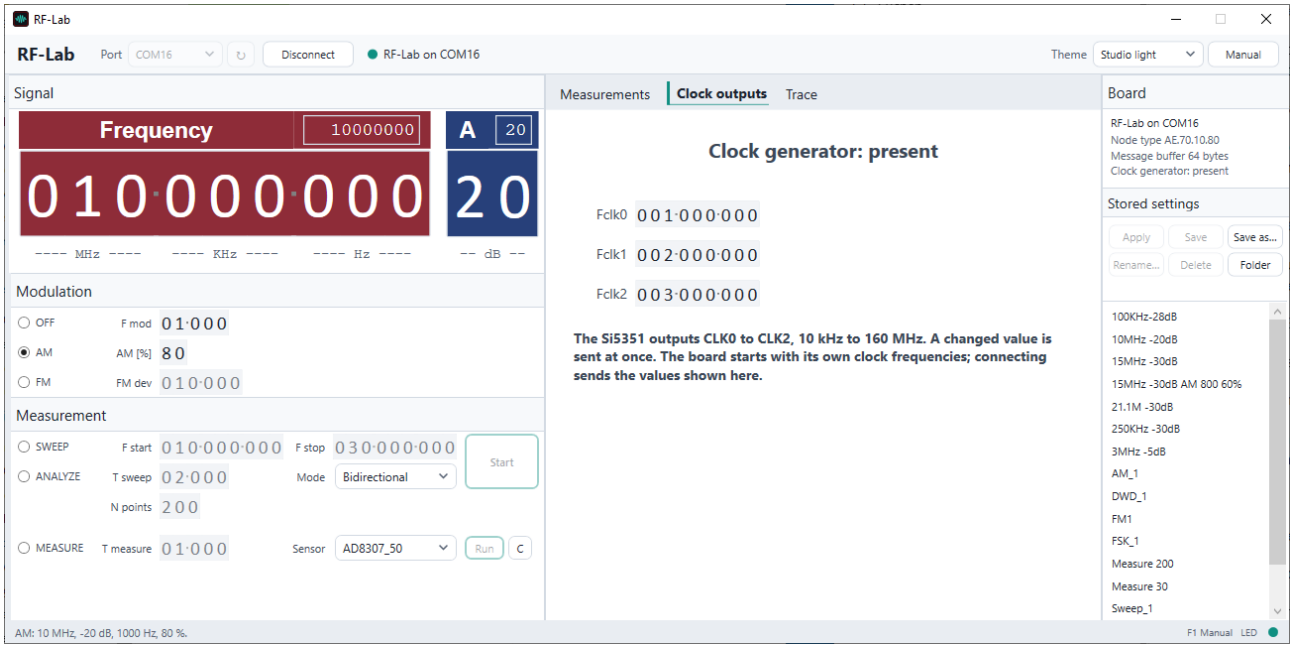
4.6 MEASURE

The level meter: the application reads the detector every **T measure** ms and plots the values; the chart starts over after 500 points, **C** clears it, **Halt** pauses and **Run** continues. The generator keeps its carrier. **Sensor** chooses the reading:

Sensor	Reading
AD8307_50	the detector's level in dB (50 Ω input): -(2235 mV − U)/25.917 − 0.5, and -70 below 200 mV
AD_8	the converter input itself in mV, 0 – 2560

The chart's dB axis reaches down to -80 dB: a quiet detector input sits at about 375 mV, which the conversion puts at about -72 dB.

5
Clock outputs



The clock outputs, Studio light theme

The Si5351 has three outputs, **Fclk0** to **Fclk2**, 10 kHz – 160 MHz each. A changed value is sent at once, to that output only. The board starts with its own clock frequencies (CLK2 at 25 MHz, for example), so the application sends the three values shown when it connects. Without a clock generator the fields stay disabled.

6
Stored settings

6.1 Using them

The list shows the files in `%LocalAppData%\RF-Lab\Presets`. A click selects a setting and shows its description; **Apply** or a double click loads it into the window and sends it to the board. Values a file does not carry keep their current value. The setting is sent as a whole, once — not value by value through intermediate states.

Key	Action
Apply	load the selected setting and send it
Save	write the window's values into the selected setting (asks first)
Save as...	store the window's values under a new name, with a description
Rename...	change the selected setting's name and description
Delete	delete the selected setting (asks first; it goes to the recycle bin)
Folder	open the folder in Explorer, for copying settings between computers

6.2 The file format

One setting is one text file, named after the setting, with one `Key,Value` line per value — the format of the original RF-Lab application, whose files work unchanged:

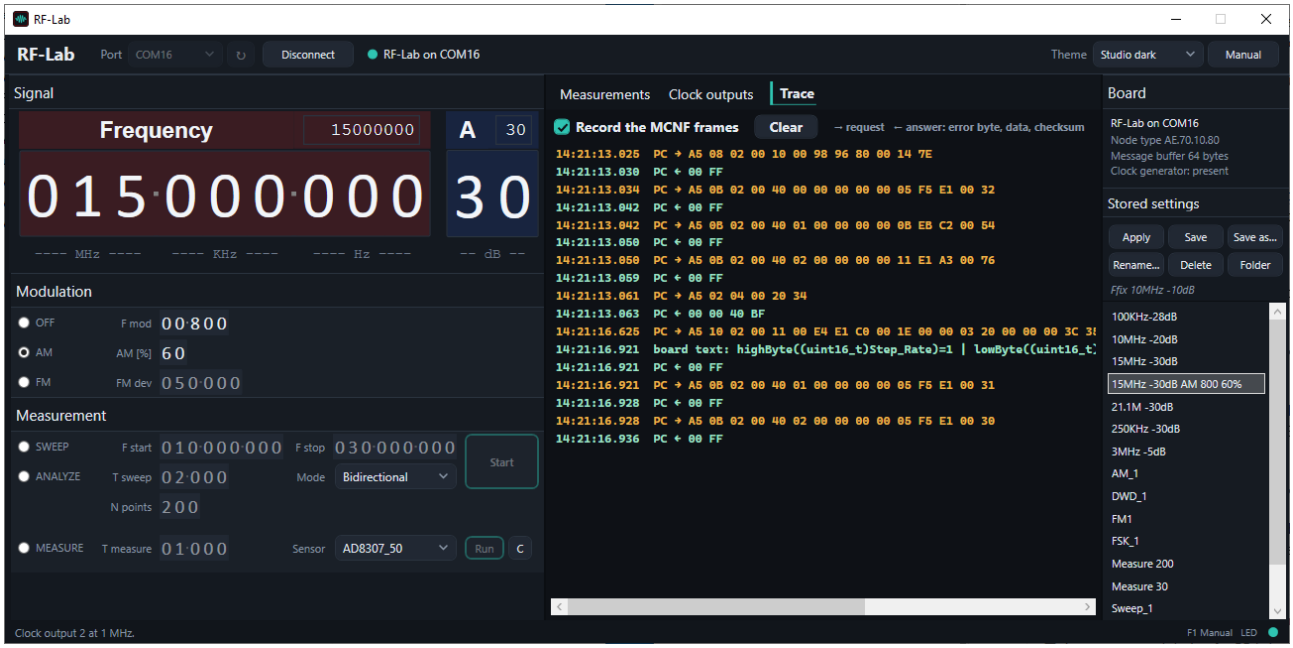
```
Name,AM_1
Description,Ffix 10MHz -10dB
Mode,1
Frequency,147500
Amplitude,40
FMod,1000
AMDepth,80
Fdev,50000
SweepMode,Ramp
Sensor,AD8307_50
Fstart,10000000
Fstop,30000000
Tsweep,10000
Tmeasure,500
Nmeasure,500
SI5351Fclk0,1000000
SI5351Fclk1,2000000
SI5351Fclk2,3000000
```

Key	Meaning
Mode	0 OFF, 1 AM, 2 FM, 3 SWEEP, 4 ANALYZE, 5 MEASURE
Frequency, Amplitude	carrier in Hz, attenuation in dB
FMod, AMDepth, Fdev	modulation frequency (Hz), AM depth (%), FM deviation (Hz)
SweepMode	Bidirectional, Ramp or SingleRamp
Sensor	AD8307_50 or AD_8
Fstart, Fstop, Tsweep	sweep span (Hz) and time (ms)
Tmeasure, Nmeasure	MEASURE interval (ms), ANALYZE points

Key	Meaning
SI5351Fclk0 … 2	clock outputs (Hz)

Keys the application does not use (the original's message generator wrote `MGENMode`, `Baudrate`, `Pattern` and others) are kept: saving a file keeps them in their place.

7
The trace



The trace: an AM setting, the firmware's text, the answer

The **Trace** tab lists every message to and from the board with its time: **PC** → a request, **PC** ← the board's answer (error byte, data, checksum). The firmware prints a few lines of debug text while it prepares AM and FM; they appear as **board text** and are not part of the answer. **Record the MCNF frames** pauses the list, **Clear** empties it. The frames are explained in `docs/Protocol.md`.

8
Themes and preferences

Classic is the original RF-Lab application's look. **Studio dark** and

Studio light are quieter: slate or white surfaces, one teal accent. The theme and the port are kept in `%LocalAppData%\RF-Lab\preferences.txt`.

`RfLab.exe` also takes arguments, for shortcuts:

Argument	Effect
`--port COM16`	chooses the port
`--connect`	connects at start
`--theme "Studio dark"`	chooses the theme

9 Scripting the board

The board speaks the MCNF protocol over its serial port; any program can

drive it. `tools/rflab.py` is a Python client (it needs pyserial):

```
python rflab.py COM16 info
python rflab.py COM16 carrier 10000000 20
python rflab.py COM16 am 10000000 20 1000 80
python rflab.py COM16 sensor 10
```

or, as a library:

```
from rflab import RfLab
with RfLab("COM16") as board:
    board.carrier(21_400_000, 40)
    print(board.sensor())
```

`docs/Protocol.md` describes every request byte by byte.

10 Troubleshooting

Symptom	Cause and remedy
"No RF-Lab board on COMx: the port is in use by another program"	another program holds the port — a second RF-Lab, a serial terminal or a HASE host. Close it.
"… the board did not answer"	the board runs another firmware, or another device is on that port. Check the port; flash the firmware.
"The node-type information does not match"	an MCNF board of another kind answers on that port (for example another HASE instrument with the same USB chip).
The port is missing	plug the board in, then □. The number follows the USB socket.
The detector reads about -72 dB whatever the generator does	the detector sees no signal: check the cable from the RF output to the detector input, and the detector's supply. A quiet AD8307 input sits at about 375 mV.
The clock outputs stay disabled	the board reports no Si5351.
The connection is closed after an error	a request was sent but no valid answer came back, so the board and the application may no longer agree. Connect again; the window's setting is sent anew.

11
Using the board with HASE

In the HASE laboratory the same board is the RF-MiniLab instrument, served

by a HASE host. Only one program can hold the serial port: stop the HASE host (or remove the board from its composition) before connecting RF-Lab, and close RF-Lab before HASE uses the board again. Nothing needs to be flashed in either direction.

Stored settings are the same files: the HASE panel reads them from `%LocalAppData%\HASE\Client\Configuration\RfLabPresets`, RF-Lab from `%LocalAppData%\RF-Lab\Presets`; copy them between the two folders.